Integration of Value Stream Map and System Dynamic

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Abstract—Value Stream Mapping (VSM) is a visual modelling tool aimed at identifying process bottle necks and lean wastes. VSM has many limitations. VSM takes a static snapshot of the value stream of a production system, or a supply chain. This research paper proposes System Dynamics approach along with VSM to overcome over this limitation of VSM. System Dynamics (SD) modelling approach is used to understand dynamics behaviour of the system. This research proposes VSM and SD integration to achieve resource optimization, reduction of process bottlenecks, Kaizen improvements etc. of the production value stream.

Keywords—Lean Manufacturing, Value Strem Map, System Dynamics

I. INTRODUCTION

Companies are under increasing pressure to streamline their value chains by reducing inventory, increasing productivity, and accelerating delivery times. Lean Manufacturing provides effective methods for identifying and eliminating waste, thereby boosting process efficiency. One key Lean technique, Value Stream Mapping (VSM), is widely used to visualize and improve production processes. However, VSM alone may not capture the complex, dynamic nature of modern production systems.

To address this limitation, integrating computer-based simulation with VSM offers a more robust understanding of both current and future production scenarios. Discrete Event Simulation (DES) is the predominant simulation method, but System Dynamics (SD) provides a broader, closed-loop perspective, particularly useful for complex production systems and supply chain analysis.

While there is considerable research on Lean and SD, few studies explore their combined use. Early work by Disney et al. integrated SD within Lean, but without a direct connection to VSM. Agyapong-Kodua et al. (2009) took a step further by incorporating SD with VSM through case studies, seeking to create a more dynamic approach to process mapping. However, a standardized method to convert VSM into quantitative SD models remains absent, hindering broader adoption.

This research aims to bridge that gap by proposing a novel framework called VSM-to-SD, designed to facilitate the translation of VSM into quantitative SD models. This framework leverages the familiar notations used in popular modeling tools like Stella® and Vensim®, offering a user-friendly process for practitioners and researchers. By enabling decision-makers to create SD-based VSMs, this framework can significantly enhance scenario analysis capabilities and support more robust decision-making, ultimately helping companies navigate the challenges of Industry 4.0 and the evolving landscape of digital transformation.

II. LITERATURE REVIEW

Lean manufacturing, originating from the Toyota Production System, focuses on minimizing waste and maximizing efficiency. A key tool in this approach is Value Stream Mapping (VSM), which helps identify value-adding activities and waste within production processes. Despite its popularity, VSM has limitations such as reliance on data accuracy and challenges in mapping complex systems.

Various methodologies have been proposed to address these limitations. Antonelli et al. (2018) suggested two simulation methods—Discrete Event Simulation (DES) and System Dynamics (SD)—for evaluating manufacturing systems. De Assis et al. (2021) introduced a VSM-to-SD framework, using tools like Stella and Vensim, to integrate VSM with SD for a more dynamic analysis. Other research, like Forno et al. (2014), explored the consequences of improper VSM use, while Kibira et al. (2009) highlighted the sustainable use of SD in different domains, including manufacturing.

Several studies proposed frameworks to enhance lean implementation. Samant and Prakash (2021) combined VSM, constraint programming, and simulation to improve lean processes. Sundar et al. (2014) presented a map for lean implementation, while Noto and Cosenz (2021) suggested that SD supports VSM in lean efforts. Additionally, Vahidi proposed using the Viable System Model for simulating VSM with SD.

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III. METHODOLOGY

The section discusses a framework for lean implementation using value stream map, system dynamics, lean box score etc. The methodology for achieving lean implementation orientation through the proposed approach is depicted in flowchart shown in Figure 1. The flowchart (point 1 to 6) is explained using the case of a production line of hot water generator product for a refrigeration equipment manufacturer located in India.

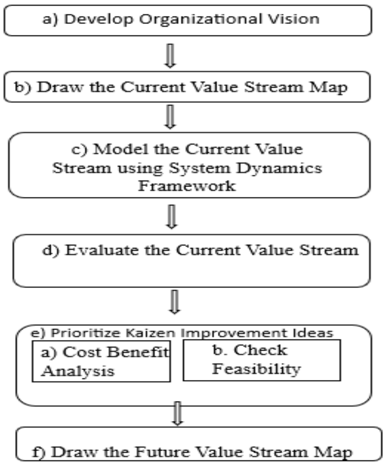

Fig. 1: Steps involved in the proposed framework.

Step a: Develop Organizational Priorities

To achieve lean implementation, it is necessary priorities of the organization are clearly defined. The priorities / goals can be externally focussed, or internally focussed (Figure 2). Externally focussed issues are those which involve carrying out changes outside the organization boundary e.g. sales, marketing issues. Internally focussed issues are those where the changes must be carried out within the boundary of the organization.

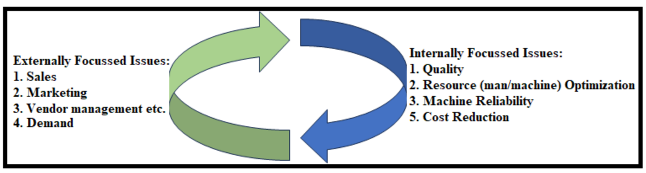

Figure 2: Organizational priorities

Step b: **Draw the Current Value Stream Map (VSM)**

Value stream map (VSM) is created for hot water production line (Figure 3). The VSM is a pictorial representation for the series of process steps involved in the production of hot water generator. The benefits of using a value stream map are that it can help to identify areas of waste and inefficiency in the production process.

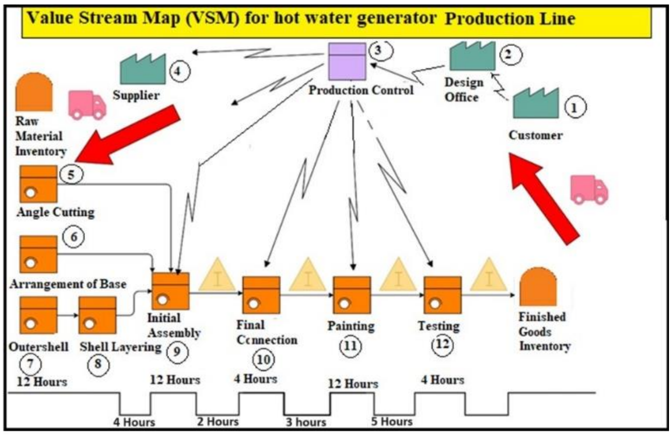

Figure 3: Value Stream Map for the hot water generator production line.

The process parameters for the various process steps of the value stream map are given in Table 2.

Table 2: Process Parameters associated with VSM

Process Step	Cycle Time (CT)	CO (Changeover time) (min)	Uptime	Quality
Angle Cutting (5)	6	30	35.8	100%
Arrangement of Base (6)	12	30	35.8	75%
Outer Shell Layers (7)	4	30	35.8	100%
Initial Assembly (9)	12	60	35.8	100%
Final Connection (10)	4	120-180	-	75%
Painting (11)	12	10	35.8	100%
Testing (12)	4	120-180	-	100%

Step c: **Model the current value stream map using systems dynamics framework.**

The System Dynamics (SD) approach has emerged as one of the very powerful computer-based tools to build simulated model complex systems to study the behaviour of the systems. The system is modelled through a combination of various sub-systems and components inter-connected through a set of flows, feedbacks, and variables. The primary building blocks of the system dynamics models are stocks, flows, feedbacks etc.

Vensim model for production system is shown in Figure 4.

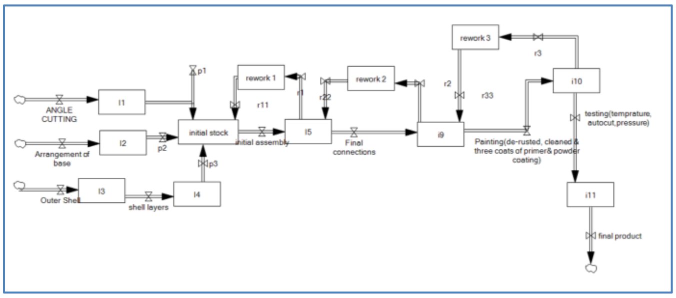

Figure 4: Vensim model of the production system

Step d: **Evaluation of current value stream mapping**

The current value stream map is evaluated using lean box score [Maskell et. al, 2011]. The method of lean box score is a comprehensive method of evaluating financial, operational, resource capacity health of the value stream. The lean box score (operational) are discussed in Table 3.

Table 3: Lean Box Score (Operational)

Operational Lean Box Score (Operational)
• Dock to Dock days: The amount of time spent from the receipt of material in the raw material warehouse to shipment of the material to the customer = 58 Hours. This is is the time taken from the receipt of raw material to material to exit the production line. • First Time Through (FTT): This parameter measures quality level in the value stream. This denotes the percentage of total products which pass through the production line without being repaired, reworked, scrapped etc. This is 75% • On-Time Delivery (OTD): Percentage of products shipped at the right time = 100% • Sales per person: Total Sales/Total number of operators employed = (Number of items produced per week) x (Price of one item)/ (Number workers employed in the value stream) = (4) x (Rs 585000)/18 = Rs 130000 • Average cost per unit: Total cost/number of units produced. Average cost per unit = (Conversion cost per week+ Inventory Cost per week +Material cost per week)/number of pieces produced in week = (87839+88+1100000)/4=Rs 296982 (Table 6)

The lean box score (resource capacity) are discussed in Table 4.

Table 4: Lean box score (Resource Capacity).

Lean Box Score (Resource Capacity)
This is the ratio of productive time and the total available time. • Paint Guns- 13% • Drilling nuts-66% • Lifting machine- 33% • Welding machine- 83% • Shearing machine-55% • Bending machine-41%

The lean box score (Financial) is discussed in Table 5.

Table 5: Lean Box Score (Financial)

Lean Box Score (Financial)
•Inventory Value (1): Cost of inventory that belongs to the value stream. $Inventory\ value = (inventory\ time\ in\ a\ week * product\ cost * 20\%)/(available\ time\ in\ week * 52)$ (14*20%*585000)/(36*52)=87.5 rupee •Revenue: Revenue from the shipped products. $Revenue = (Number\ of\ pieces\ per\ year * product\ cost)/52$=2,340,000 rupee •Material cost (2): $Material\ cost = (Number\ of\ pieces\ produced\ per\ year * Raw\ material\ cost)/52$=1,100,000 rupee •Conversion Costs (3), i.e Costs incurred in converting the raw material. = Rs 87839 (Table 6). Total Costs = (1) + (2) + (3) Value stream profit = Revenue – Costs = Rs 2049215

The conversions costs have been discussed in Table 6:

Table 6: Explanation of Conversion Cost

Machine name	Ownership Cost (weekly) (Rs) (A)	Running Cost (weekly) (Rs) (B)	Manpower Cost (Rs) (C)	Conversion Cost (Rs) (D=(A)+(B)+(C)
Paint gun	1923	1521	11250	14694
Drilling nut	769	60	3750	4579
Lifting machine	11538	9087	7500	28126
Welding machine	673	11200	7500	19373
Shearing machine	2885	170	7500	10555
Bending machine	2885	128	7500	10512
	20673	22166	45000	87839

Step e: **Kaizen improvement**

Kaizen improvement is a process of systematic improvement, where the lean practitioner studies the value stream in detail, and comes up with continuous improvement. These improvement efforts can come up in a variety of areas e.g. quality, reliability, productivity, supply chain, training, manpower optimization etc. These Kaizen improvement efforts are discussed in Table 7

Table 7: Problems and Kaizens

Problem Description, and Kaizen Improvement
Problem: Metal painting is poor adhesion of the paint to the surface. Kaizen Improvement: Cleaning of metal with sandpaper and metal wire brush This will save rework cost Rs 600.
Problem: Rust Formation; If the metal surface is not properly protected, rust can form underneath the paint. Kaizen Improvement: Remove any existing rust by sanding or using a rust remover before painting, saving cost Rs 600.
Problem: Supply Chain Disruptions Delays or shortages in the supply chain can halt production on the assembly line. Kaizen Improvement: Maintain strong relationships with suppliers and implement inventory management systems.
Equipment and Machinery Breakdown: Kaizen Improvement: Implement regular maintenance schedules.

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IV. RESULTS AND DISCUSSION

In a study on improving the production line of a hot water generator, Value Stream Mapping (VSM) and System Dynamics (SD) were integrated to identify and resolve bottlenecks. VSM provided a high-level view of the process, while SD offered insights into the system's dynamics and interactions. This combined approach, called VSM-to-SD, helped uncover critical gaps in areas like painting, rust formation, assembly, equipment malfunction, and personnel training. Kaizen improvements were implemented to address these issues, including pre-painting metal cleaning to reduce rework costs, enhanced inventory management to avoid supply chain disruptions, regular equipment maintenance, and comprehensive staff training. Manpower optimization was also achieved through takt time analysis, reducing weekly labor costs by Rs. 104,000. This collaborative methodology demonstrates a practical framework for improving complex production systems.

V. CONCLUSION

Industry 4.0 leaders use decision-support systems to analyze value stream improvements. Value Stream Mapping (VSM) is a proven tool, but simulations (like VENSIM) offer more dynamic future insights. This paper proposes a VSM-to-SD framework, which simplifies System Dynamics for VSM users. The results empower faster decision-making with minimal gaps and can be applied in various production settings. This framework can also bridge the gap for VSM experts with limited modeling experience, promoting wider adoption.

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